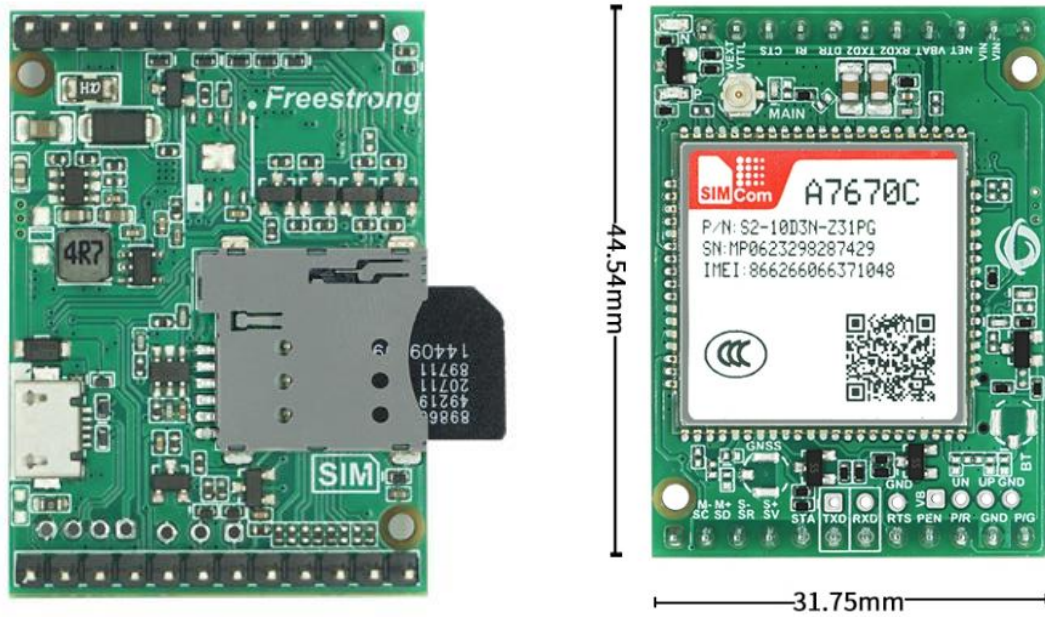


# FSC24&24 double row function board

## specification sheet

| system | Model number     | system | Model number  | system | Model number      |
|--------|------------------|--------|---------------|--------|-------------------|
| GSM    | SIM800A-L        | LPWA   | SIM7070E      | LTE    | A7670C-FANS       |
|        | SIM800A32-L      |        | SIM7070E      |        | A7670C-FASC       |
|        | SIM800F-L        |        | SIM7070E-REEL |        | A7672S-LASE       |
|        | SIM800F-L        |        | SIM7070E-MN   |        | A7672S-FASE       |
| LPWA   | SIM7000C         | LPWA   | SIM7070G-HP   |        | A7670C-GW         |
|        | SIM7000C         |        | SIM7070E      |        | A7670G-LLSE       |
|        | SIM7000E         |        | SIM7070G-HP   |        | A7670C-LASE-PCIEA |
|        | SIM7000E         |        | SIM7070G-M    |        | A7670C-LASS-PCIEA |
|        | SIM7000A         | LTE    | A7670C-BASL   |        | A7670E-LASE-PCIEA |
|        | SIM7000A         |        | A7670C-BASS   |        | A7670G-LABE       |
|        | SIM7070E-NG      |        | A7670E-FASE   |        | A7670C-LANV       |
|        | SIM7000JC        |        | A7670E-LASE   |        | A7670C-LANS       |
|        | SIM7000JC        |        | A7670E-LASC   |        | A7670C-MANV       |
|        | SIM7000G         |        | A7670E-MASC   |        | A7670C-MANS       |
|        | SIM7070G-NG      |        | A7670SA-FASE  |        | A7670C-LASE       |
|        | SIM7070G-NG-REEL |        | A7670SA-LASE  |        | A7670C-LASC       |
|        | SIM7070G         |        | A7670SA-LASC  |        | A7670C-LASL       |
|        | SIM7070G-REEL    |        | A7670SA-MASC  |        | A7670C-LASS       |
|        | SIM7070G-PCIE    |        | A7670C-FASL   |        | A7670C-MASL       |
|        | SIM7070G-MN      |        | A7670C-FASS   |        | A7670C-MASS       |

## ★Core Board Size:



| Pin number (P1) | NAME | Pin description                                                                                                                                                                                     | Remark                                                                         |
|-----------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1               | SC   | Analog voice MIC_N                                                                                                                                                                                  |                                                                                |
| 2               | SD   | Analog voice MIC_P                                                                                                                                                                                  |                                                                                |
| 3               | SR   | Analog voice SPK_N                                                                                                                                                                                  |                                                                                |
| 4               | SV   | Analog voice SPK_P                                                                                                                                                                                  |                                                                                |
| 5               | STA  | Power-on status indication. When the core board is powered on and initialization is completed, STA outputs high level, otherwise it remains low level.                                              | Used to determine whether the core board is powered on;<br>Voltage range: 3.3V |
| 6               | TXD  | Main serial port sending, used for AT command configuration and data transmission                                                                                                                   | Voltage range: 3.3V;<br>Can be enabled by VTTL changes to other levels         |
| 7               | RXD  | Main serial port reception, used for AT command configuration and data transmission.                                                                                                                |                                                                                |
| 8               | RTS  | Hardware flow control, main serial port RTS input                                                                                                                                                   | Voltage domain 3.3V                                                            |
| 9               | PEN  | Core board power enable pin. By default, internal pull up to VIN, power on automatically. Pull down the hour core,the power supply of the core board has no output. If not in use, it is suspended. |                                                                                |
| 10              | P/R  | Module power on/off/reset control, default is PWK pin, can be changed to RST pin                                                                                                                    |                                                                                |
| 11              | GND  | Negative pole of power input                                                                                                                                                                        |                                                                                |
| 12              | P/G  | Default is GND, can be changed to VIN                                                                                                                                                               |                                                                                |
| Pin number (P2) | NAME | Instructions                                                                                                                                                                                        | Remark                                                                         |
| 13              | VEXT | Internal 1.8V power output, maximum output current 50mA, cannot power high-power loads                                                                                                              | Static sensitive, no need to float                                             |

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|                        |             |                                                                                                                                                                       |                                          |
|------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 14                     | VTTL        | Level conversion circuit external reference level input, invalid by default                                                                                           |                                          |
| 15                     | CTS         | Hardware flow control, main serial port CTS output                                                                                                                    | Voltage domain 3.3V                      |
| 16                     | RI          | Ringing indication, outputs low level for 60ms when receiving URC report, and then returns to high level state                                                        | Voltage domain 3.3V                      |
| 17                     | DTR         | Sleep wake-up, when the core board enters sleep mode, pull DTR low to wake up the core board                                                                          | Voltage domain 3.3V                      |
| 18                     | TXD2        | Serial port 2 is sent, AT firmware is not used, just leave it floating.                                                                                               | Voltage domain 3.3V                      |
| 19                     | RXD2        | Serial port 2 receives, AT firmware is not used, just leave it floating                                                                                               | Voltage domain 3.3V                      |
| 20                     | VBAT        | Module power input, supports 3.4~4.2V input, typical value is 3.8V, can be used for lithium battery power supply power; when using VIN power supply, VBAT output 3.8V | PEN pin is disabled when powered by VBAT |
| 21                     | NET         | Network status indication                                                                                                                                             |                                          |
| 22                     | VIN         | Positive power input, supports 5-16V DC input                                                                                                                         | You can connect one or both.             |
| 23                     | VIN         |                                                                                                                                                                       |                                          |
|                        |             |                                                                                                                                                                       |                                          |
| <b>Pin number (P3)</b> | <b>NAME</b> | <b>Instructions</b>                                                                                                                                                   | <b>remark</b>                            |
| 24                     | TXD         | The main serial port is sent to facilitate monitoring the data interaction between the main control and the core board after the core board is installed.             | for auxiliary testing                    |
| 25                     | RXD         | The main serial port receives to facilitate monitoring the data interaction between the main control and the core board after the core board is installed.            |                                          |
| 26                     | GND         | Main serial port auxiliary test common ground pin                                                                                                                     |                                          |
| 27                     | VB          | USB_VBUS, VBUS presence detection input                                                                                                                               | USB XH2.0 terminal block                 |
| 28                     | UN          | USB_N, USB bus differential negative pole                                                                                                                             |                                          |
| 29                     | UP          | USB_P, USB bus differential positive                                                                                                                                  |                                          |
| 30                     | GND         | GND, ground                                                                                                                                                           |                                          |

注意:

Notice:

1. For conventional applications, only VIN, GND, TXD, RXD, and PEN are needed. Other pins can be left floating if they are not used.
2. VBAT and VIN cannot be used as power supply inputs at the same time. When VBAT is used to power the core board, the PEN pin is disabled.

# Application interface

## 1 Power supply

The core board has three power supply modes: VIN, VBAT, and USB.

VIN: The power supply range is 5V~16V, the power needs to be more than 8W, and the recommended power supply is 5V 2A.

VBAT: The power supply range is 3.3V~4.3V, generally used for lithium battery power supply.

USB: Powered by 5V via USB cable. Some computer USB interfaces (recommended to connect to computer USB 3.0) have insufficient driving capabilities.

The core board may not boot or restart frequently. At this time, you need to connect VIN to power the core board.

Notice:

1. VIN and VBAT cannot power the core board at the same time.
2. The actual input voltage must be within the minimum and maximum range.

## 2 Switch on and off

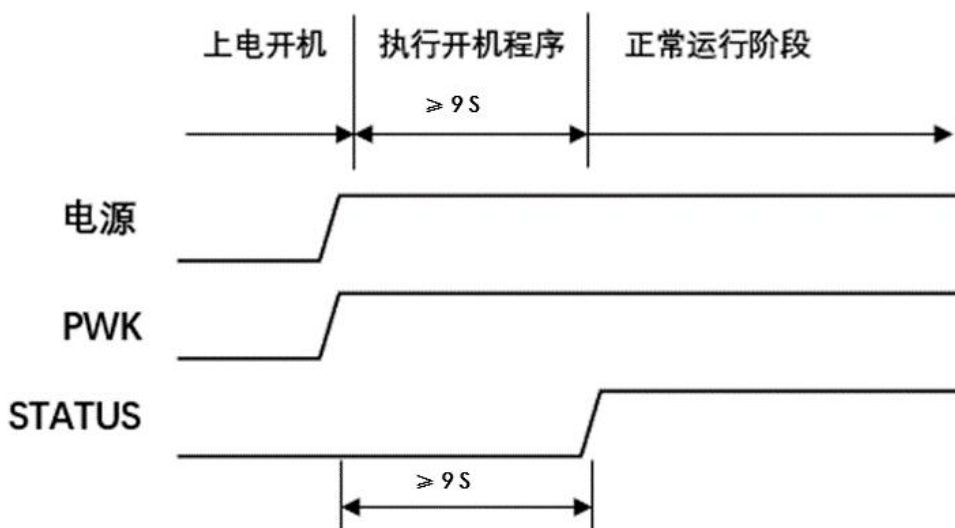
The core board automatically powers on by default. In applications, it is recommended to use the PWK and PEN pins to control the power on and off of the core board in order to reduce power consumption and handle abnormal situations.

| 应用场景                         | PWK                  | PEN                            |
|------------------------------|----------------------|--------------------------------|
| 设备长连接, 无开关机需求                | 可悬空                  | 建议启用, 当遇到异常时, 控制 PEN 强制断电重启    |
| 有低功耗需求/不需要长连接的应用, 需要控制核心板开关机 | 启用, 通过主控 IO 控制核心板开关机 | 建议启用, 但 PWK 失效时, 控制 PEN 强制断电重启 |

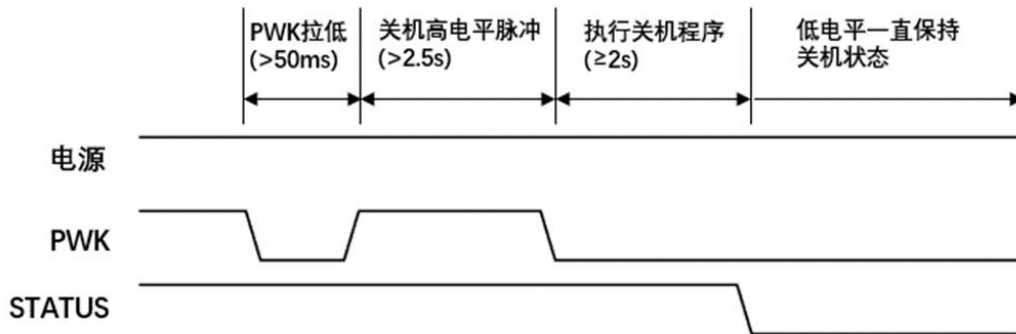
Note: Under normal circumstances, use the PWK pin to shut down. Only when PWK fails, pull the PEN pin low to disconnect the power supply of the core board to shut down. It is not recommended to shut down the computer frequently through the PEN pin, because long-term abnormal operation may cause the file system in the Flash to be damaged, causing the core board to fail to boot normally.

### 2.1 PWK

Power-on: The core board automatically powers on by default. The power-on sequence of PWK is shown in the power-on sequence diagram:



**Shutdown:** In the power-on state, pull the PWK pin low for more than 50ms and then pull it high for at least 650ms. The core board enters the shutdown program. At this time, the PWK pin is pulled low and kept for more than 2S to ensure the complete execution of the shutdown program (after the shutdown program is completed, keep The low-level core board will remain in the shutdown state), and the shutdown timing is as shown in the PWK shutdown timing diagram:



Notice:

When the PWK pin fails to shut down, PEN can be controlled to directly cut off the power supply to achieve shutdown;

The voltage domain of the PWK pin is 3.3V.

## 2.2 PEN

When the PWK pin fails to shut down, PEN can be controlled to directly cut off the power supply to achieve shutdown. Control PEN using MCU' s GPIO

When using the pin, it is recommended to add a transistor driver.

**Power on:** By default, PEN is internally pulled up to VIN, that is, the core board's default power supply is enabled, and it is powered on.

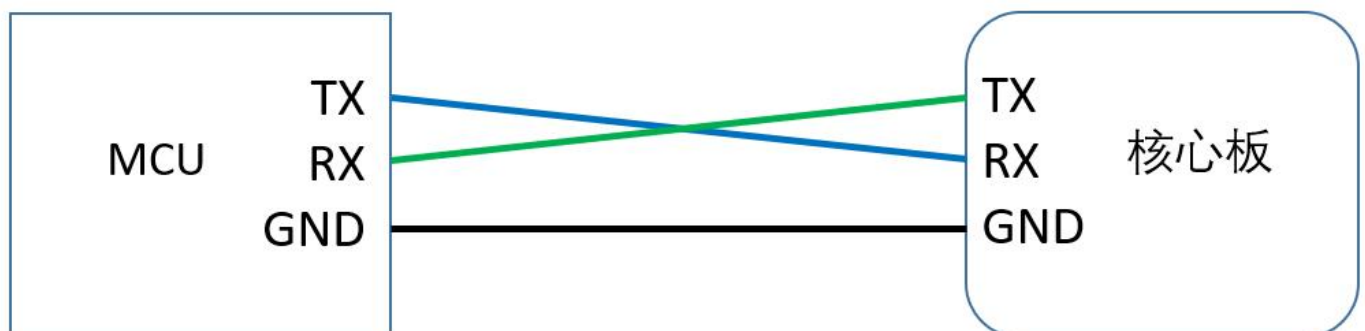
**Shutdown:** Pull the PEN pin low to disconnect the power supply of the core board to achieve power-off shutdown.

## 3 TTL Serial port

The core board has 2 serial ports, serial port 1 and serial port 2, of which serial port 1 is the main serial port and is used for AT command interaction and data transmission. Serial port 2 has no effect in AT firmware and can be left floating.

Supported baud rates are: 4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800bps, etc.

TXD and RXD are cross-wired when communicating with the master.



The default serial port is 3.3V TTL level. If the user's main control serial port level is not 3.3V, level matching needs to be done and the core can be modified.

jumper resistor on the board and connect the VTTL pin to the target voltage.

| IO 电平值 | 更改说明                         | 备注            |
|--------|------------------------------|---------------|
| 3.3V   | 焊接 RT1, 不焊接 RT2              | 板子出货默认焊接方式    |
| 其他电压   | 焊接 RT2, 不焊接 RT1, VTTL 接到目标电压 | 需要自行更改或联系我司销售 |

Notice:

After modifying the IO level, the levels of the TXD, RXD, TXD2, RXD2, DTR, RI, CTS, RTS, and STA pins will all change to VTTL.

### 3.1 DTR awakening

The DTR pin is the sleep mode trigger pin of the core board and can be left floating when not in use; entering sleep mode can significantly reduce standby power consumption, and the main serial port function will be turned off in sleep mode. When the host and core board are connected through the serial port, the module can enter sleep mode through the following steps: Use the AT+CSCLK=1 command to enable the sleep function. Make sure DTR remains high (default internal pull-up is high).

The module can be woken up by the host pulling DTR low (it needs to be kept low in sleep mode to keep the core board awake).

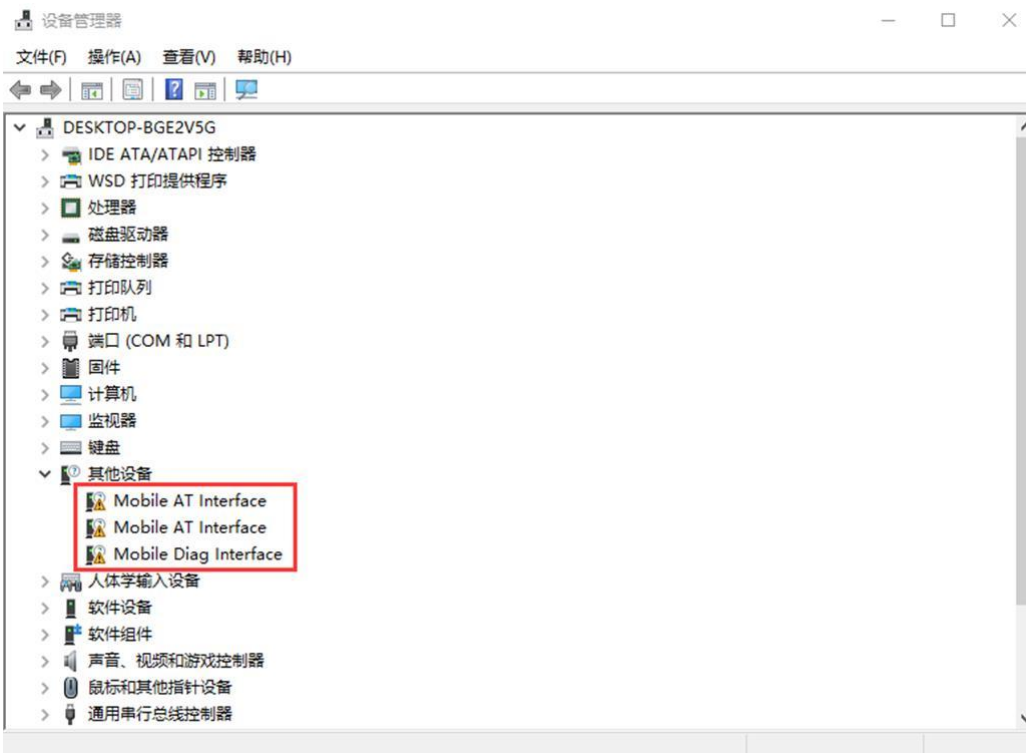
## 4 USB interface

Notice:

1. The USB cable must have data transfer function.
2. Communication using the USB interface requires the installation of a USB driver.

After the core board is connected to the computer via USB, if the driver is not installed successfully, open the computer device manager and click "Other Devices"

If you see three devices with yellow exclamation marks, just update the drivers accordingly



### PC side test method

Precautions:

Precautions:

The serial port cable needs to be cross-connected, that is, the TX of the core board is connected to the RX of USB to TTL, and the RX of the core board is connected to the TX of USB to TTL.

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The peak power of the core board can reach 8W. The USB power supply capability of some computers cannot meet this requirement. The core board may frequently restart due to insufficient power supply (shown as the serial port always outputting power-on information). In this case, it is recommended to use an independent power supply to power the core board. The core board supports 5V~16V power input. During debugging, a "common ground" is required, that is, the GNDs of the core board, power supply, and USB to TTL are connected together.

After power on, the core board power indicator light (red light) is always on. Wait for about 10 seconds, and the network indicator light (blue light) starts to flash. At this time, it means that the core board is powered on and can debug AT commands.

surface Indicator light status description

|                                      | Indicator status     | Module working status                              |
|--------------------------------------|----------------------|----------------------------------------------------|
| Power indicator light (red light)    | Steady on            | Power-on                                           |
|                                      | Extinguish           | Power off, shut down                               |
| Network indicator light (blue light) | Steady on            | Search status                                      |
|                                      | 200 ms on/200 ms off | Data connection established, or network registered |
|                                      | Extinguish           | Module shutdown                                    |

| procedure | status | AT command and return value | Instructions                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|--------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | send   | AT                          | Handshake test.                                                                                                                                                                                                                                                                                                                                                                                                |
|           | return | OK                          | Returning "OK" means that the serial port communication is normal.                                                                                                                                                                                                                                                                                                                                             |
| 2         | send   | AT+CPIN?                    | Check SIM card status.                                                                                                                                                                                                                                                                                                                                                                                         |
|           | return | +CPIN: READY<br>OK          | Returning "READY" indicates that the card reading is successful. If ERROR is returned, the card reading fails.                                                                                                                                                                                                                                                                                                 |
| 3         | send   | AT+CSQ                      | Check the RF signal quality.                                                                                                                                                                                                                                                                                                                                                                                   |
|           | return | +CSQ: 24,99<br>OK           | Parameter 1: Signal quality (0~31), make sure the signal quality is greater than 18, otherwise the data communication may be unstable.<br>0 - 113dBm or below<br>1-111dBm<br>2..30 - 109dBm... -53dBm<br>31 -51dBm or higher<br>99 Unknown or undetectable<br>Parameter 2: Not enabled and can be ignored.                                                                                                     |
| 4         | send   | AT+CEREG?                   | Check network registration status.                                                                                                                                                                                                                                                                                                                                                                             |
|           | return | +CEREG: 0,1<br>OK           | Parameter 1: Controls unsolicited result codes, 0 is disabled.<br>Parameter 2: Registration status, 1 means registration is successful.<br>0: Not registered, the module has not yet searched for a new operator to register. 1 Registration successful<br>2: Not registered, but the module is searching for a new operator to register with 3 Registration refused<br>4: Unknown<br>5: Registration, Roaming |